

FELIX DOUVEN

Email me: felix@quaternionic.net
Website: quaternionic.net

LinkedIn: linkedin.com/in/felixdouven
GitHub: github.com/Flexiis

PROFESSIONAL EXPERIENCE

ASML – Competence Engineer, Electrical Qualification; Eindhoven, NL Sep 2026 – Present

- Co-own electrical qualification of EUV and DUV actuators (HV DC, partial discharge, resistivity, LCR and gravity-compensation tests); currently standardizing first-pass yield tracking
- Lead a cycle-time reduction project using live non-linear regression models on production data

ASML – Technical Support Engineer, Twinscan Manufacturing; Eindhoven, NL Sep 2024 – Sep 2026

- Raised production's in-house problem-solving rate from 88% to 98% through sensor and electrical troubleshooting knowledge transfers
- Led 4 factory-wide escalations: 2 solved structurally (material-induced weakness); glue-adhesion failure traced to a new tool; 1 software-caused system down fixed via 8D, triggering a supplier change
- Cut non-value-added time in new product introduction (NPI) products by replacing a process step (Lean, SPC), saving ~€500 per disturbance at ~6 disturbances a week (~€150K a year)
- Saved ~140 hours a year by automating production workflows in Python and analyzing production data with Pandas & Matplotlib

VSParticle – Prototype Engineer Intern, Electrocatalysis; Delft, NL Nov 2023 – Apr 2024

- Built a working gas recycling prototype for a nanoparticle printer, raising gas reuse from 0% to 88% and saving up to ~€70K per unit per year
- Automated the test campaign in Python via the printer's API: back-to-back 20-min runs at 0/40/80% recycle ratios with automatic gas flushing and 1 Hz sensor logging to a database for analysis
- Designed an Arduino PID pressure regulator (proportional valve + sensor) holding 1200 mbar within ± 8 mbar; showed an 80% cut in carrier-gas use is feasible without altering the printer's design

Philips – Prototype Engineer Intern, Innovation (Medical Systems); Best, NL Feb 2022 – Jul 2022

- Designed the 7-axis upper robot arm for a 1:1 mockup of the IGT system, a robotic feasibility study for the ceiling-mounted X-ray C-arm; the mockup was used to secure early-stage stakeholder funding
- Compared 4 wrist drivetrain concepts against a 1.88 kg weight budget; chose sheet metal for manufacturability and sized parts with FEM, deflection calculations and a MATLAB DH model

EDUCATION

Fontys University of Applied Sciences | B.Eng. Mechatronics Engineering | Eindhoven, NL | Sep 2019 – Apr 2024

Specialization: Adaptive Automated Systems

- Control, electrical and mechanical engineering; systems engineering (V-model, UML); robot arm control (C++)

Chuo University | Exchange Semester, Minor in Global Management | Tokyo, JP | Apr 2023 – Sep 2023

- Holland Scholarship recipient; 9 courses incl. Finance, Change Management, International Cooperation, Public Policy and Japanese; 8 of 9 graded A or S (highest grade)

PROJECTS

Autonomous Underwater Vehicle – Electrical Lead & System Designer; client: NOX Robotics Sep 2022 – Jan 2023

- Led the electrical design and defined the system architecture in a 7-person team, redesigning the AUV for 30 m depth (was 6 m); built the 14.8 V Li-Po power system (BMS, e-stop) and a leak-detection and sensor PCB

Custom 3D Printer – Personal Project 2024 – 2025

- Designed and built a 3D printer from scratch in SolidWorks (aluminium frame, MGN15 linear rails, dual-Y belt drive, E3D V6 hotend); iterated a custom fan duct through 3 versions; prints 80 mm/s on Marlin 2.0

Housing Market Aggregator – Personal Project 2026

- Built a scraper for ~32 Dutch housing sites feeding PostgreSQL, with a FastAPI back end and map-based web UI; self-hosted with Docker on a home server

Sensor Data Classification – Fontys course project 2022

- Recorded our own phone-sensor dataset and reached 95.85% activity-classification accuracy by combining KNN, Random Forest and Extra Trees

SKILLS & CERTIFICATIONS

Certifications: SolidWorks Professional (CSWP, 2022) – Mechanical Design & Sheet Metal; SolidWorks Associate (CSWA, 2019); Lean Manufacturing Yellow Belt (ASML); Kepner-Tregoe Problem Solving

Engineering: System design, circuit design, FEM analysis, control systems, SPC, DMAIC, root cause analysis

Software: SolidWorks CAD, MATLAB, Python, C++, Siemens NX, TIA Portal (PLC), PCCSIM, Docker, SQL

Languages: Dutch (native), English (fluent), Japanese (basic)